



Swing Trader · After-Fee Workflow

For a reader who trades nothing and has trained no model. Five columns, one per stage, read left to right. **Is there a rule that clears the fee, on data it has never seen?** Every stage answers a piece of that, and *every check here was written without opening its file*.

WORKFLOW CHEAT SHEET

seamusrobertmurphy/swing-trader
Crypto NO-GO · Equities live on paper
State of 7 September 2026

A The bar

what every idea must clear

A1 Transaction costs

0.20% ROUND TRIP

A round trip pays a fee on entry and exit, so a shorter bar interval pays it more often. The 5-minute frames carries positive gross expectancy; the round-trip cost is six times that.



```
# the argument, one line  
kept = move - round trip - slippage  
allow if kept ≥ 1.5%
```

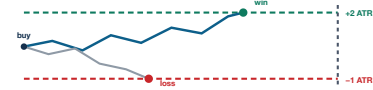
	0.20	0.15	1.5
% assumed round trip			
% best case round trip			
% the least worst have			

ROUND-TRIP COST EXCEEDS THE MEASURED EDGE A basis point is a hundredth of one per cent. The best crypto edge is five to twelve of them *smaller* than the fifteen to twenty a round trip costs. Nothing tried has closed it.

A2 Label geometry

TRIPLE BARRIER

The label is a modelling decision, not an observed quantity. Three barriers are set at entry: a take-profit, a stop, and a holding-period limit. Whichever the path touches first determines the class, so the same price series yields different labels under different barrier geometry.



Both lines are set in units of the coin's own daily travel, so a calm coin and a violent one are judged on equal terms.

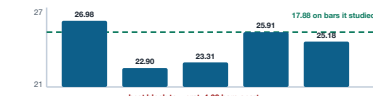
BARRIER EXPECTANCY IS NEGATIVE BEFORE FITTING On the 4-hour panel the profit line is reached first on **254** bars in every thousand. Aiming twice as far up as down, you need **333** to come out level. The arithmetic is losing before anything is fitted, so the first thing to check is always the target's own record, not the model's.

A3 Acceptance criteria

60% OF FOLDS

A pooled total can be carried by a single favourable regime, so the sample is partitioned into half-year folds and the edge must hold in **60 per cent** of them, on two criteria.

did it pay did the picks make money after fees
did it beat did they beat the average name



THE BAR AT WORK, SHARE MOMENTUM, 5 SEP 2026



FOLD DISPERSION, NOT THE MEAN Those five blocks average to what a steady 24.9 would. A steady 24.9 would be usable. These are not.

TRAPS THAT COST DAYS

every one of these happened here, and each was found late

Figures quoted without opening the record

Three figures on the previous version of this sheet were wrong because they were recalled rather than read. Open the file that produced the number, every time, including your own.

A render can succeed and still be wrong

A page can build, exit cleanly, and report 0 of 6 cycles where there are 3. Compare the new render against the last one before believing it.

B The data

sources, screening, ranking

B1 Data sources

TWO ARCHIVES

Two static archives rather than live feeds. Both were downloaded once and read from disk, so the build is deterministic and reproducible.

Crypto. Binance's own monthly files of open, high, low, close and volume, one zip per coin per month, checksum-verified.

Shares. Alpaca daily bars, split and dividend adjusted. Of 12,57 live US companies, **2,660** clear \$20m a day above \$3 a share.

PANEL	ROWS	NAMES	INPUTS
5 min	5.53M	11	96
1 hour	1.64M	47	61
4 hour	3.94M	567	90
1 day	0.52M	540	104
shares 1d	5.00M	2,547	96

DELISTED PAIRS RETAINED IN THE ARCHIVE **612** pairs ever traded against **487** alive today. Build the list from today's exchange and you delete every coin that failed, which makes any result an upper bound. The 1-hour panel is still the old 47-coin build.

B2 Point-in-time screen

FOUR CRITERIA

Four criteria, applied conjunctively. Membership is computed at every bar from information available at that bar, so the screen is point-in-time rather than retrospective.

DOOR	BAR IT HAS TO CLEAR
Trading enough	≥ 30m USDt in 24h
Moving enough	ATR 2.5–12% of price
Cheap to cross	≤ 0.5% estimated
Old enough	≥ 120 days lived

ATR, the average true range, is how far price usually travels in a day, as a share of price. The band shown is daily; a 4-hour bar wants 1.0 to 4.9%.

The **spread**, the difference between the best bid and best ask, is paid on entry and exit. The archive carries no order book, so it is estimated from each day's high and low by the Corwin-Schultz estimator and gated at 0.5%. The live screen reads the quoted spread and gates at 0.05%.

SCORED ACROSS THE MARKET, TRADED ON A SUBSET The full universe is ranked; only assets with positive out-of-sample contribution are admitted.

B3 Cross-sectional rank

SURVIVING SIGNAL

Time-series direction, the probability an asset rises, failed on every frame. **Cross-sectional rank**, the ordering of assets each bar, did not.



SIGNAL CONFIRMED, EXECUTION RULE REJECTED The ranking held across train and test. The one execution rule built on it, gated on a rising bitcoin trend, passed 27 per cent of folds against 60. Untested: longer holds, and the same ranking on equities.

TERCILE WIDTH APPROXIMATES THE UNIVERSE Measured 8 September: **266** assets at the median bar, so the top tercile is **89** names, on a stale comment claiming 5 to 7 per bar.

C The signals

what the model is given

C1 Feature families

FIVE BLOCKS

THE COIN'S OWN PRICE HISTORY
day-length windows
short windows
chart indicators
WHICH WAY THE TREND POINTS
Supertrend
adaptive version
calm or wild
WHAT THE REST OF THE MARKET IS DOING
bitcoin, leading
slower bars above
crowding
WHO WAS IN A HURRY TO TRADE
buyers against sellers
order-book texture

RAW LAGS MATCH THE ENGINEERED FEATURE SET Asked how many bars a trend has left, the 90 built inputs miss by **24.213** bars on the 4-hour panel. Twenty-four plain past returns, nothing built at all, miss by **24.247**. Guessing the average misses by 26.477. The two real answers sit a thirty-fourth of a bar apart, so they hold the same information and a memory-based model has no gap left to close.

C2 Feature evidence

90 FEATURES

Ninety features, most carrying no weight. **alone** is univariate AUC less 0.5; **steady** the share of five folds preserving sign; **used** permutation importance on the Brier score.

FAMILY	N	ALONE	STEADY	USED
vs bitcoin	7	.019	.83	.0069
pandas-ta	7	.049	1.00	.0023
adaptive Supertrend	6	.008	.60	.0004
TA-Lib extras	15	.012	.75	.0003

RELATIVE STRENGTH DOMINATES OWN-PRICE FEATURES The strongest family by a factor of three is strength against bitcoin, led by **F.BTC_MOM_168**, a 168-hour relative move. The only family not derived from the asset's own price.

ZERO PERMUTATION IMPORTANCE, UNSTABLE SIGN Eight of the ten unread columns are candlestick patterns. **16 OF 90** barely move the score when permuted; **14 OF 90** reverse sign between folds.

C3 Indicator engines

FOUR INDICATORS

Four indicators on one OHLCV series, each reduced at every bar to a discrete stance. The confluence score is the count in agreement.

READ	WHAT FIRES IT
MACD	guarded crossover, carried as a regime
Moving average	fast line above or below slow
Fibonacci	price reacting at a retracement level
Candles	bullish or bearish engulfing

CROSSOVER SUPPRESSION NEAR THE ZERO LINE Crossovers near zero fire on noise. A cross is suppressed unless it clears a band set at half the rolling standard deviation of the histogram, then holds sign for a further bar.

MACD is the 12-period less the 26-period exponential moving average, against its own 9-period average.

ILLUSTRATIVE, NOT A TRADING SIGNAL Built on crypto for a closed strategy, run unchanged on 66 US symbols. Nothing trades until it clears lane A.

D The fitting

splits, selection, tuning

D1 Train and test split

CHRONOLOGICAL

Random partitioning places later observations in training and earlier ones in test. Returns are autocorrelated, so this leaks future information and inflates every score.



The yellow sliver is the **gap**. Without it the last rows studied still await outcomes landing inside the test window.

SINGLE SCORING OF THE HELD-OUT YEAR The final year is opened at the end and scored once. A setting chosen by looking at it has spent it, and it cannot be made unseen.

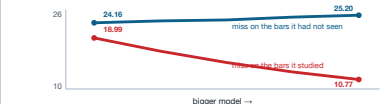
AS CONFIGURED

PIECE	SETTING
Held back	the final 365 days
Studied	all history before it
Gap at the cut	8 days (48 4h bars)

D2 Hyperparameter sweep

NINE SETTINGS

Nine hyperparameter combinations, each scored on three walk-forward folds of the 4-hour panel, target being bars until the Supertrend reverses.



The miss on studied bars nearly halves while the miss on unseen bars gets worse. All nine settings fail the 1.1 rule, and the one that came top was the least powerful in the grid.

HOW FAR EACH MOVES THE SCORE, IN BARS all nine settings changing the fold

FOLD VARIANCE EXCEEDS CONFIGURATION VARIANCE A separate comparison of three feature representations spanned **0.094** bars. This is the argument for walk-forward scoring over a single split, and against ranking configurations on one fold.

D3 Variable selection

ELASTIC NET

Ninety features against the available sample size admits overfitting. Retention is decided by an elastic-net penalty on the training split, before the model is fitted.

penalty every coefficient is pushed toward zero; uselessness reach it

strength chosen by cross-validation, then loosened by one standard error

kept whatever still has a coefficient after that
Taking the loosest setting within one standard error of the best, rather than the best itself, deliberately keeps fewer inputs than the score alone would. The best-scoring setting is usually the one that has begun fitting the noise.

SELECTION CONFINED TO THE TRAINING SPLIT Choosing inputs is a decision, so it spends the data it is made on. Made on the blind year, it would spend that year, and the year cannot be made unseen again.

AFTER-FEE OUT-OF-SAMPLE SCORE DECIDES A penalty says which inputs a regression keeps. It does not say whether a family earns its place. That is settled by C2 and by the after-fee score in lane F.

E The verdict

results and the live book

E1 Model assessment

TWO DEFECTS

The model emits a probability, so the error is the distance between that probability and the realised outcome.

ratio unseen miss divided by studied miss

MODEL	STUDIED	UNSEEN	RATIO
HistGBM	.2619	.4818	1.840
LightGBM	.2886	.4826	1.720
Random forest	.4526	.4840	1.069
Logistic	.4855	.4909	1.011

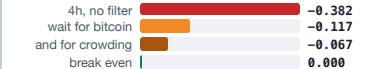
1-hour panel, 6 Sep 2026. A ratio near 1 generalises. Above 1.1 it has memorised, and is thrown out.

TWO DEFECTS IN MODEL SELECTION AND CALIBRATION The winner was chosen on the unseen miss alone, never the ratio, crowning HistGBM at **1.840** where the rule rejects past 1.1. And every model treats rare wins as common, pushing its probabilities away from what this score measures: on 4-hour bars it misses by **0.4844** where answering "25.4 per cent, always" misses **0.4353**. Nothing checks whether a stated 30 per cent happens 30 per cent of the time.

E2 Scoreboard

AFTER FEES

Net expectancy per crypto trade after fees, held out. Gates raised it without reaching zero.



SHARE MOMENTUM, PER CENT A MONTH, 116 MONTHS top tenth average share

NEGATIVE EXPECTANCY GROSS OF COST Fee excluded: daily **-1.150%**, 4-hour **-0.182%**. Only 5-minute is positive at **+0.033%**, six times under the round trip.

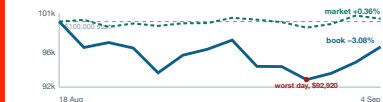
THE ONE THAT WORKS

12-1 equity momentum. Rank on the prior twelve-month return excluding the latest month, hold the top decile. Spread **2.41** standard errors against a bar of two, positive in 85 per cent of folds, above market in 80.

Rejected: 1-hour direction, AUC 0.50; entry sharpening, 29% of folds; breadth gate, 27%.

E3 Live paper book

LIVE PRICES



Fourteen daily closes against holdings at Friday's prices, \$96,916.58 from \$100,000. The plan expects an 8.2% swing; a 3.1% dip is four tenths of it.

49 shares held at 1.8% each	3/6 clean weekly cycles done	7.6 bp a fill mid-session
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weekly, never at the opening bell
alpaca trade, no rebalance, execute

PAPER ACCOUNT, LIVE PRICES A paper account at real prices; every order path stops unless it reads the exact word true. Fixed in code: never short, never borrow, never options, never add to a loser, 5% most in one name, 10% least in cash.